

# Empirical Proof Record: INCONCLUSIVE

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## Empirical Proof Record — INCONCLUSIVE

**Proof ID:** Odd21782212c19b5f1ae2cebb887ddf9...

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### 1. Claim

Across a long single-pass stream, the substrate still recalls earlier experiences beyond a fixed context window: (for every probe whose lag exceeds `window_ref=256`, recall Jaccard  $\geq 0.50$ ). The reported floor is the worst beyond-window probe.

- **Operationalisation:** Stream 8 distinct stimuli once through Kimera's entity target, then re-probe 8 positions spread over the stream. For each probe, `recall = Jaccard(concept set at first exposure, concept set at probe)`; `lag = intervening cycles`. `recall_floor_beyond_window = min recall over probes with lag > 256`; the invariant holds iff that floor  $\geq 0.50$ .
- **Threshold:** `recall_floor_beyond_window >= 0.5 jaccard`
- **H0:** H0: `recall < 0.50` for some item recalled beyond `window_ref=256` — the horizon is shorter than the reference window
- **H1:** H1: `recall  $\geq 0.50$`  for every beyond-window probe — memory reaches past a fixed window of that size

### 2. Verdict

**INCONCLUSIVE** — observed 0.0

recall floor 0.0000 over 0 beyond-window probes (lag > 256); mean recall 0.0000, stdev 0.0000; memory horizon (max lag with recall  $\geq 0.50$ ) = 8 cycles; max lag measured 8; 8-item single-pass stream; 0 probes produced no concept set; too few beyond-window probes (<3) to decide — widen the stream or lower `window_ref`

### 3. Evidence

| Pillar                | Statistic                  | Value | 95% CI | Library         |
|-----------------------|----------------------------|-------|--------|-----------------|
| 0.flow.memory_horizon | recall_floor_beyond_window | 0.0   | —      | ophamin 0.115.2 |

### 4. Pre-registration

- Registered at: 2026-05-23T23:11:25.978419+00:00
- Config hash: 97758cc1fa8320dc018877a25a8d436e...
- Data hash: 221a8a035c85eee67ce8bccf6670a709...

### 5. Signature

70219d4947950ccf3e93411cc1c0783d4204ae1d61502affd4ba3888334485c3